Experiment 8

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Title: Implementation and Comparative Analysis of SARSA and Q-Learning

Software Required:

- Python
- Gymnasium
- Matplotlib

AIM

To implement SARSA and Q-Learning and compare their learning performance.

THEORY

SARSA is an on-policy algorithm that updates Q-values using the action actually selected by the current policy. Q-Learning is an off-policy algorithm that uses the maximum estimated future Q-value.

PROCEDURE

1. Open the Python environment.
2. Import Gymnasium and Matplotlib.
3. Create the required environment.
4. Initialize separate Q-tables for SARSA and Q-Learning.
5. Set the learning rate, discount factor and exploration rate.
6. Train the agent using the SARSA algorithm.
7. Train the agent using the Q-Learning algorithm.
8. Record the rewards obtained during each episode.
9. Compare the learning performance of both algorithms.
10. Plot and analyze the SARSA and Q-Learning results.

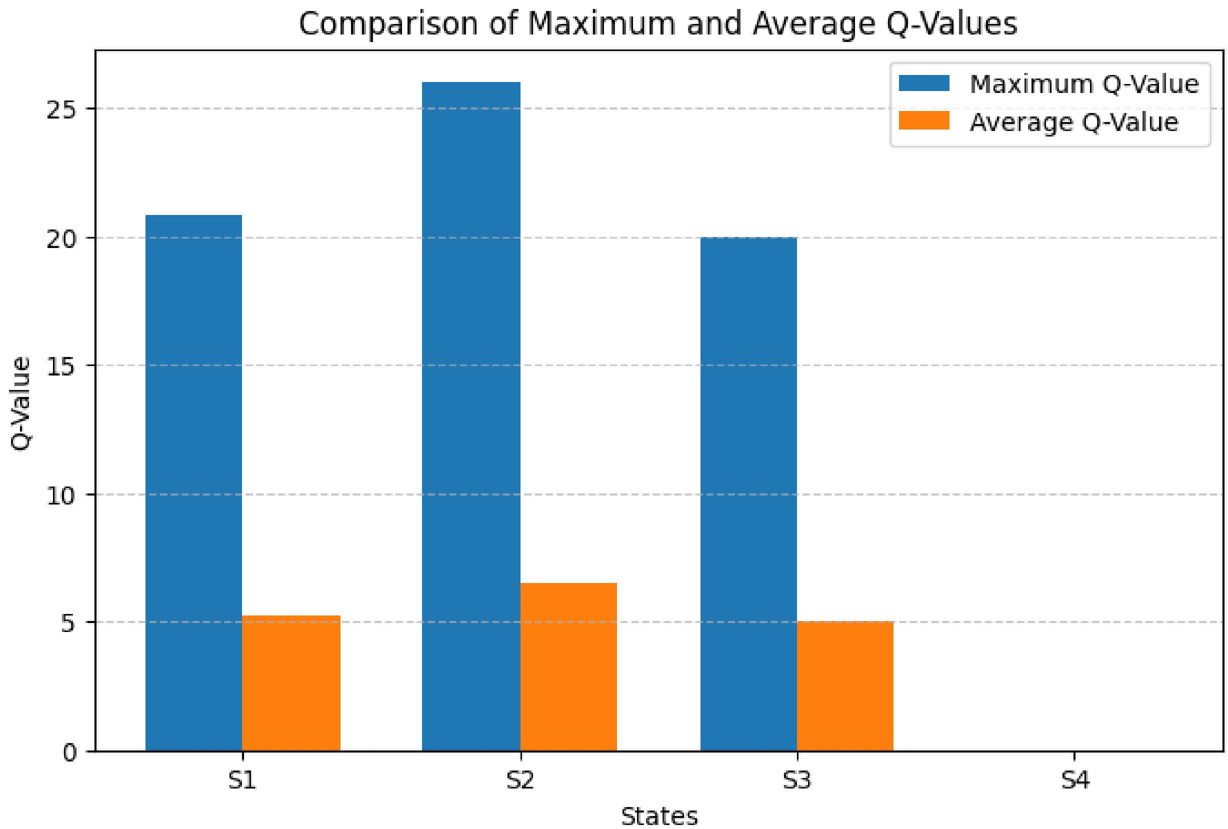
CODE:

```
import numpy as np
import matplotlib.pyplot as plt

# Reward Matrix
R = np.array([
    [-1, 0, -1, -1],
    [-1, -1, 10, -1],
    [-1, -1, -1, 20],
    [-1, -1, -1, 0]
])

gamma = 0.8
alpha = 0.9
episodes = 1000

states = 4
Q = np.zeros((states, states))

np.random.seed(42)

# Q-Learning
for episode in range(episodes):
    state = np.random.randint(0, states)

    while state != 3:
        actions = np.where(R[state] >= 0)[0]
        action = np.random.choice(actions)
        next_state = action

        Q[state, action] += alpha * (
            R[state, action] +
            gamma * np.max(Q[next_state]) -
            Q[state, action]
        )

        state = next_state

print("Final Q-Table\n")
```

```python
print(np.round(Q, 2))

# Comparison Data
max_q = np.max(Q, axis=1)
avg_q = np.mean(Q, axis=1)

states_name = ['S1', 'S2', 'S3', 'S4']
x = np.arange(len(states_name))
width = 0.35

# Comparison Graph
plt.figure(figsize=(8,5))

plt.bar(x - width/2, max_q, width, label='Maximum Q-Value')
plt.bar(x + width/2, avg_q, width, label='Average Q-Value')

plt.xticks(x, states_name)
plt.xlabel("States")
plt.ylabel("Q-Value")
plt.title("Comparison of Maximum and Average Q-Values")
plt.legend()
plt.grid(axis='y', linestyle='--', alpha=0.7)

plt.show()
```

OUTPUT:

```
Final Q-Table

[[ 0.   20.8  0.    0. ]
 [ 0.    0.  26.    0. ]
 [ 0.    0.   0.   20. ]
 [ 0.    0.   0.    0. ]]
```



RESULT

SARSA and Q-Learning were successfully implemented and their learning performances were comparatively analyzed.